

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 07

Expectation of DRV

Def: Let X be a discrete RV with PMF $f_X(\cdot)$ and support S_X . The expectation or mean of X is defined by

$$E(X) = \sum_{x \in S_X} x f_X(x) \quad \text{provided} \quad \sum_{x \in S_X} |x| f_X(x) < \infty.$$

- ▶ $E(X)$ is the weighted average of the values taken by X .
- ▶ If $\sum_{x \in S_X} |x| f_X(x) = \infty$ then we say that expectation does not exist.

Example 1: X = outcome of a roll of a fair die. What is $E(X)$?

Example 2: $X \sim \text{Bin}(n, p)$. What is $E(X)$?

Example 3: $X \sim \text{Geo}(p)$. What is $E(X)$?

Example 4: $X \sim \text{Poi}(\lambda)$. What is $E(X)$?

Example 5:

$$f_X(x) = \begin{cases} \frac{c}{n^2}, & x \in \mathbb{N}, \quad \text{where } c = \left(\sum_{n=1}^{\infty} \frac{1}{n^2} \right)^{-1} \\ 0 & \text{otherwise.} \end{cases}$$

Let X be a DRV having the above PMF, then $E(X)$ does not exist.

Expectation of CRV

Def: Let X be a CRV with PDF $f_X(\cdot)$. The expectation of X is defined by

$$E(X) = \int_{-\infty}^{\infty} xf_X(x)dx \quad \text{provided} \quad \int_{-\infty}^{\infty} |x|f_X(x)dx < \infty.$$

Example 6: $X \sim U(a, b)$, what is $E(X)$?

Example 7: $X \sim \text{Exp}(\lambda)$, what is $E(X)$?

Example 8: $X \sim N(\mu, \sigma^2)$, what is $E(X)$?

Example 9: Let X be a CRV having PDF $f_X(x) = \frac{1}{\pi(1+x^2)}, \forall x \in \mathbb{R}$.

What is $E(X)$?